

The Listening City System: A Smart Citizen Feedback Platform for Urban Incident Reporting

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Abstract. The Listening City System is a smart citizen feedback platform designed to support urban incident reporting and response management. The system allows citizens to submit reports about public issues such as traffic accidents, fire incidents, damaged infrastructure, environmental problems, and security concerns. Each report may include a description, GPS location, images, or videos to help authorities verify and process the issue more effectively.

The platform supports different user roles, including citizens, ward staff, and police staff. Citizens can create reports, track processing status, receive notifications, and provide feedback after the issue is handled. Ward staff and police staff can receive assigned reports, update processing results, and coordinate responses based on incident type and location. The system aims to improve communication between citizens and local authorities, reduce manual reporting procedures, and increase transparency in urban management.

Keywords: Smart City · Citizen Feedback · Urban Incident Reporting · GPS · Public Service Management

1 Introduction

The rapid development of smart city technologies has transformed the way governments interact with citizens and manage urban services. Modern cities increasingly rely on digital platforms to improve public service delivery, enhance transparency, and encourage citizen participation. One of the key challenges faced by local authorities is the efficient collection and management of citizen feedback regarding public infrastructure, environmental issues, traffic violations, and security incidents.

Traditional reporting methods, such as telephone hotlines, paper forms, and in-person visits, often suffer from slow response times, limited tracking capabilities, and inefficient communication between citizens and government agencies. As a result, many urban issues remain unresolved for extended periods, reducing citizen trust in public administration.

To address these challenges, this project proposes City Connect, a Smart City Citizen Reporting System designed to facilitate communication between citizens,

ward authorities, police officers, and city administrators. The platform enables citizens to submit incident reports with GPS location information, communicate with authorities through real-time chat, and track the processing status of their reports using a unique tracking code. Furthermore, the system integrates AI-assisted emergency support and OCR-based license plate recognition to improve public safety management.

The objective of City Connect is to provide a centralized platform that improves urban governance, increases citizen engagement, and enhances the efficiency of issue resolution processes. The proposed solution supports multiple user roles, including Citizen, Ward Staff, Police Officer, and Super Administrator, ensuring clear responsibility assignment and effective collaboration among stakeholders.

2 Problem Statement

In many cities, citizens may not know the correct department to contact when an incident occurs. Reports can be delayed, duplicated, or sent to the wrong authority. In addition, local staff may have difficulty verifying whether a report is real or fake if there is not enough supporting evidence.

Therefore, a centralized feedback platform is needed to collect citizen reports, classify incident types, assign reports to suitable authorities, and provide transparent status updates.

3 Related Work

Citizen participation has become an essential component of smart city development. Previous studies have emphasized that smart cities should not only focus on technology adoption but also provide mechanisms for citizens to actively contribute to urban governance. Citizen engagement platforms allow residents to report problems, provide suggestions, and participate in decision-making processes.

Several smart city initiatives have implemented digital service request systems that enable citizens to report urban issues through web or mobile applications. These systems typically support complaint submission, issue categorization, status tracking, and communication with local authorities. Research has shown that such platforms can improve service transparency and reduce response time.

In addition, recent studies have explored the concept of citizen-centric smart cities, where technology is used to strengthen the relationship between citizens and government agencies. Smart governance frameworks emphasize the importance of collecting citizen feedback, monitoring public service performance, and providing timely responses to community concerns.

Despite these advancements, many existing solutions focus primarily on issue reporting and lack integrated features such as real-time communication, intelli-

gent assistance, police collaboration, and automated incident analysis. Furthermore, some systems provide limited support for assigning responsibilities among different government departments.

The proposed City Connect platform extends existing approaches by combining citizen reporting, real-time communication, AI-assisted support, OCR-based license plate extraction, and administrative monitoring within a unified system. This integrated approach enables more efficient collaboration between citizens and local authorities while improving transparency and accountability throughout the incident management process.

4 System Analysis and Design

The main actors of the system include Citizen, Ward Staff, and Police Staff.

Citizens can register, log in, submit incident reports, upload images or videos, track report status, and give feedback after the issue is resolved.

Ward Staff can receive reports related to local public issues, verify information, update processing status, and provide final results.

Police Staff can handle urgent or security-related incidents such as fire alerts, accidents, and public safety problems.

5 Method

3.1 System Development Approach

The development of City Connect follows a requirement-driven system analysis and design methodology. The system requirements were identified based on common urban governance challenges and the need for efficient citizen-government communication. The development process consists of requirement analysis, use case modeling, database design, user interface design, and deployment planning.

3.2 Requirement Analysis

The system is designed to support four primary actors:

Citizen Ward Staff Police Officer Super Administrator

Citizens can register and authenticate using OTP, submit incident reports, automatically capture location information through Google Maps integration, and communicate with the AI emergency assistant. Ward Staff members receive reports within their administrative area, communicate with citizens through real-time chat, and update investigation results. Police officers can process security-related incidents and utilize OCR technology to extract vehicle license plate information from uploaded images. Super Administrators monitor city-wide performance through dashboards and KPI reports.

3.3 Use Case Design

The Use Case Diagram defines the interactions between system actors and functional components.

Citizen Functions Register and Login using OTP Submit Incident Reports Automatically Detect Location via Google Maps Chat with AI Emergency Assistant Ward Staff Functions Receive Urban Reports Provide Real-Time Citizen

Support Update Incident Resolution Results Police Functions Extract License Plate Information Using OCR Super Administrator Functions Monitor Real-Time Dashboards and KPI Statistics

The use case model ensures clear role-based access control and separates responsibilities among stakeholders.

3.4 Database Design

The database architecture is represented through an Entity Relationship Diagram (ERD) consisting of five major entities:

USERS

Stores user authentication and authorization information, including username, encrypted password, role, and account status.

WARDS

Represents administrative areas responsible for managing citizen reports.

CATEGORIES

Stores classifications of incident reports, such as Infrastructure, Security, Environment, and Traffic.

FEEDBACKS

Serves as the core entity of the system. Each feedback record contains a tracking code, title, description, GPS coordinates, address details, processing status, assigned staff member, OCR results, and timestamps.

CHAT MESSAGES

Stores communication messages exchanged between citizens and authorities within a feedback case.

The relationships between entities support efficient management of reports, user assignments, category classification, and communication history.

3.5 User Interface Design

The user interface is designed following accessibility and usability principles. The homepage provides access to major system functions and includes accessibility features such as adjustable font sizes for elderly users.

The feedback management interface presents incident reports through searchable lists, category filters, and interactive maps displaying report locations. KPI dashboards provide visual summaries of total reports, resolved incidents, pending cases, and average response times. These visualizations assist administrators in monitoring system performance and identifying service bottlenecks.

3.6 Deployment Architecture

The system utilizes MySQL as the relational database management system. To simplify deployment and maintenance, all services are containerized using Docker. The deployment environment is configured through a docker-compose.yml file, enabling consistent execution across development, testing, and production environments.

Docker-based deployment improves portability, scalability, and maintainability while reducing environment configuration issues.

6 Proposed Solution

The proposed system provides a web-based platform where citizens can submit reports with incident type, description, GPS location, and media evidence. The system stores report information in a database and supports report assignment based on category and location.

For urgent incidents, the system can provide a fast reporting flow so that important information is sent quickly to the responsible authority. To reduce fake reports, the system may require user authentication, OTP verification, evidence upload, and report history tracking.

7 Implementation Plan

The system can be developed using a front-end and back-end architecture. The front-end provides user interfaces for citizens and staff. The back-end handles authentication, report management, assignment workflow, notification, and database operations.

The database stores user accounts, report categories, incident reports, locations, media files, status history, and feedback records.

8 Conclusion

The Listening City System aims to improve communication between citizens and local authorities by providing a digital platform for urban incident reporting. The system helps citizens report problems faster and allows authorities to manage reports more transparently and efficiently. In the future, the system can be extended with AI-based report classification, dashboard analytics, and integration with emergency services.

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